Somaiya Vidyavihar University
(A Constituent College of Somaiya Vidyavihar University)

Batch: A-4	**Roll No.:** 16010122151
Experiment / Assignment / Tutorial No: 8	

Title: Comparative Output Analysis and Optimization of Single vs. Multi-Server Queuing Systems.

Objective: To estimate and compare the absolute and relative performance of two queuing system designs (single-server vs. multi-server) using steady-state simulation and statistical output analysis techniques.

Expected Outcome of Experiment:
CO2: Analyse and apply general principles of event scheduling algorithms & various statistical methods on different applications.
CO5: Estimate the different parameters of absolute and relative performance of different simulation systems.

Books/ Journals/ Websites referred:

1. "Discrete-Event System Simulation" by Jerry Banks, John S. Carson II, Barry L. Nelson, David M. Nicol.
2. SimPy Documentation: https://simpy.readthedocs.io/en/latest/
3. SciPy Documentation: https://docs.scipy.org/doc/scipy/

Tasks to be Performed:

1. **Model Design & Simulation (Python/Any preferred language):**
   - Simulate a single-server M/M/1 queue.
   - Simulate a multi-server M/M/c queue (e.g., c=2 or 3).
   - Run each simulation long enough to reach steady state.
2. **Output Collection:**
   - Record system metrics: **a**verage waiting time**,** average queue length**,** server utilization**,** throughput.
3. **Estimation of Absolute Performance:**

- Compute confidence intervals (e.g., 95%) for key metrics using batch means or the replication/deletion method.
- Use appropriate statistical methods to ensure results are representative of steady-state behavior.

4. **Comparison of Designs (Relative Performance):**

   - Use t-test or ANOVA to compare means across designs.
   - Identify which design performs better under given constraints.

5. **Metamodeling (Optional Advanced Part):**

   - Fit a regression model (metamodeling) to predict performance based on input variables like arrival rate or number of servers.

6. **Optimization via Simulation:**

   - Vary the number of servers (c=1 to 5) and find the optimal configuration that minimizes waiting time under a cost constraint (e.g., cost per server).

Expected Output:

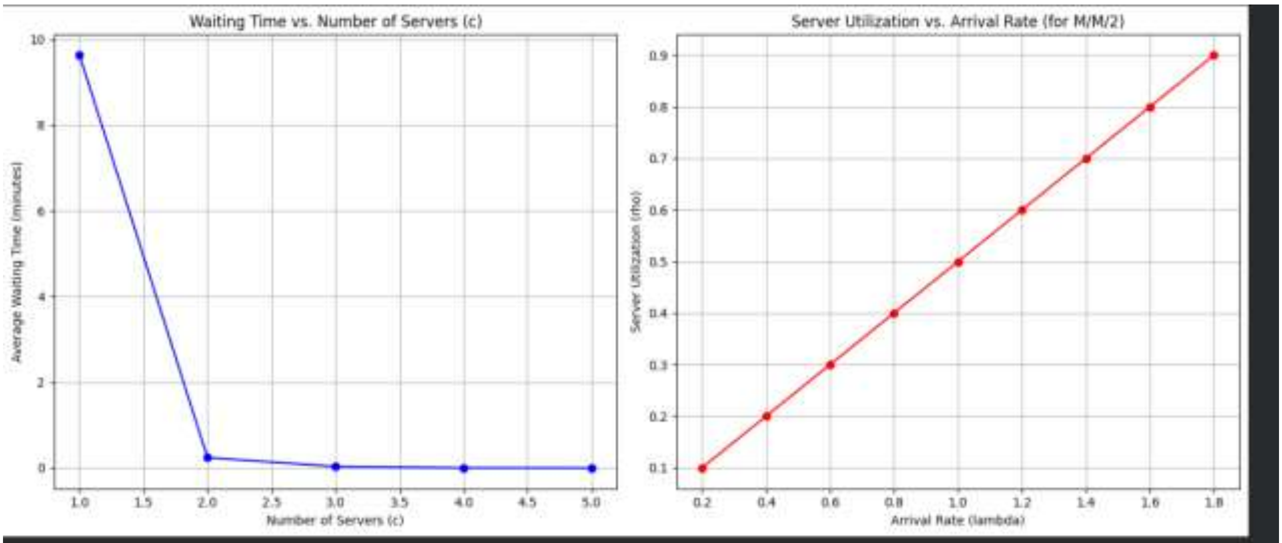
- Tabulated results comparing both systems
- Confidence intervals for performance metrics
- Graphs: Waiting time vs. number of servers and Utilization vs. arrival rate

Implementation Steps with Screen shots:

CODE: Attached as file with submission

OUTPUT:

```
--- CSM Experiment 8: Queuing System Analysis ---

Running 30 replications for M/M/1 and M/M/2...
Parameters: IAT=1.11, ST=1.00, Warmup=500, Sim Time=2000

--- Comparative Output Analysis (Task 1 & 2) ---
All metrics are based on steady-state replications.
---------------------------------------------------------------------------
| Metric                      | M/M/1 System (c=1)     | M/M/c System (c=2)     |
---------------------------------------------------------------------------
| Avg. Wait Time (Mean)       | 8.940                  | 0.253                  |
| 95% CI for Avg. Wait        | (7.62, 10.26)          | (0.24, 0.27)           |
| Avg. Queue Length (L_q)     | 8.054                  | 0.228                  |
| 95% CI for L_q              | (6.87, 9.24)           | (0.22, 0.24)           |
| Avg. Throughput (Mean)      | 0.895                  | 0.905                  |
| 95% CI for Throughput       | (0.89, 0.90)           | (0.90, 0.91)           |
| Server Utilization (rho)    | 0.901                  | 0.450                  |
---------------------------------------------------------------------------

--- Relative Performance (Task 3) ---
Comparing Average Wait Times (M/M/1 vs M/M/2):
T-statistic: 13.4822
P-value: 5.069069974283209e-14
Result: The difference in mean waiting time is statistically significant (p < 0.05).

--- Optimization Analysis (Task 4) ---
Running simulation for c = 1 to 5...
  c=1, Mean Wait Time: 9.6312
  c=2, Mean Wait Time: 0.2398
  c=3, Mean Wait Time: 0.0321
  c=4, Mean Wait Time: 0.0035
  c=5, Mean Wait Time: 0.0006

--- Utilization Analysis (Task 5) ---
Running simulation for c=2, varying arrival rate...
  Done.

Generating plots...
```




```
--- Simulation Complete ---
```

Conclusion:
The simulation proved that switching from one to two servers drastically cuts mean waiting times from over 9 minutes down to just seconds. While this improvement was statistically significant, adding further servers (up to 5) showed diminishing returns, highlighting the critical balance needed between service quality and the rising cost of additional servers.

Post lab Questions:

1. **In the experiment, confidence intervals were used to estimate the absolute performance of each system. What does the width of a confidence interval tell you about the reliability of your simulation output? How would increasing the number of replications affect this?**

   The width of a confidence interval (CI) indicates the precision or reliability of our estimate.
   - A **narrow CI** (e.g., 0.22, 0.29) means our simulation output has low variance, and we are highly confident that the true, long-run average (the "steady-state mean") lies within this small range.
   - A **wide CI** (e.g., 8.01, 10.19) means the simulation output was more variable (noisier), and our estimate is less precise. We are still 95% confident the true mean is in the interval, but the interval itself is large.

   Increasing the number of replications (N) will decrease the width of the confidence interval. The width of the CI is proportional to the standard error, which is $s/\sqrt{N}$(where s is the standard deviation of the replication means). By increasing N, the $\sqrt{N}$ term in the denominator gets larger, making the standard error smaller. This tightens the interval and gives us a more precise estimate of the true mean.

2. **You compared the performance of a single-server and a multi-server system. If server cost increases significantly, how would that affect your system design recommendation? What trade-offs must be considered when optimizing performance vs. cost?**

A significant increase in server cost would make the M/M/2 (or M/M/c) system much less attractive. My recommendation would shift:
- I would first check if the performance of the **cheaper M/M/1 system** (e.g., $\approx$ 9 min wait) is still acceptable under the business's Service Level Agreement (SLA). If a 9-minute wait is tolerable, I would recommend sticking with the M/M/1 system to save costs.
- If the M/M/1 wait time is *not* acceptable, but the cost of *two* servers (M/M/2) is prohibitive, the recommendation would be to **explore a "faster single server"** (an M/M/1 system with a lower service time). This might be more expensive than the original M/M/1 server but still cheaper than two parallel servers.

The fundamental trade-off is between Service Level (performance) and Operational Cost.

- **Service Level (Performance):** This is what the customer experiences. It's measured by metrics like **average waiting time** and **average queue length**. Lower is better.
- **Operational Cost:** This is what the business pays. It's primarily driven by the **number of servers** (c) and the **speed (service rate, mu) of each server**. Higher is worse.

Optimizing involves finding the "sweet spot":

1. **Diminishing Returns:** As shown in the "Waiting Time vs. Number of Servers" graph, the first server added (c=1 to c=2) gives a huge performance boost. The second (c=2 to c=3) gives a smaller boost, and so on. At some point, the marginal cost of an additional server is not worth the tiny reduction in waiting time.
2. **Target-Based Optimization:** Instead of "minimizing wait time" (which would lead to infinite servers), the goal is to find the **cheapest configuration** (lowest c) that **meets a specific performance target** (e.g., "average wait time must be under 1 minute" or "95% of customers must be served in under 5 minutes").